

Mutual Fund Industry Analytics: A Comprehensive Data Engineering and Financial Analytics Capstone Project

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1. Executive Summary

The Indian mutual fund industry has witnessed remarkable growth over the past decade, with Assets Under Management (AUM) crossing ₹50 lakh crore in 2024. Despite this growth, retail investors often lack access to rigorous, data-driven tools to evaluate fund performance, manage risk, and make informed investment decisions. Asset Management Companies (AMCs), similarly, face challenges in understanding investor behaviour, predicting SIP churn, and benchmarking fund performance against peers.

This capstone project, titled **Mutual Fund Industry Analytics**, addresses these challenges through a full-stack data analytics solution built entirely in Python and SQLite. The project ingests, cleans, models, and analyses six categories of structured financial datasets covering 40 mutual fund schemes across 10 leading AMCs, spanning a timeline from 2022 to 2025. The pipeline processes over 46,000 NAV records and 32,778 investor transaction records, computes institutional-grade performance metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown, and delivers findings through an interactive Tableau dashboard and a rule-based fund recommender system.

Key outcomes of the project include:

- A production-ready ETL pipeline with automated data ingestion, validation, and SQLite persistence via a star-schema data model
- A multi-metric Fund Scorecard ranking 40 schemes across five weighted dimensions
- Discovery that SBI Mutual Fund leads industry AUM at ₹84,91,000 Cr, followed by ICICI Prudential MF (₹62,93,000 Cr) and HDFC Mutual Fund (₹57,32,000 Cr)
- A critical finding that approximately **97.8% of SIP investors** are classified as "At Risk" based on irregular payment gaps, signalling a systemic retention crisis
- A Historical VaR and CVaR analysis providing tail-risk quantification at the 95th confidence level for all 40 funds
- An interactive Tableau dashboard visualising industry KPIs, AUM trends (2022–2025), and AMC-level rankings

The project demonstrates the application of data engineering, financial analytics, and business intelligence techniques to a high-impact industry domain, and is positioned as a replicable framework for AMC analytics teams and fintech organisations.

2. Problem Statement

2.1 Industry Context

The Indian mutual fund ecosystem, governed by SEBI and represented by AMFI, manages assets across equity, debt, hybrid, and thematic fund categories. With 26.12 crore folios as of the analysis period and total SIP inflows of ₹9,39,721 Cr, the industry is at an inflection point where data analytics can drive both investor outcomes and business competitiveness.

2.2 Core Challenges

Despite scale, the industry faces several structural data challenges:

For Investors: Information asymmetry is pervasive. Retail investors typically evaluate funds based on trailing returns alone, without understanding risk-adjusted performance, downside exposure, or how a fund behaves relative to market benchmarks. This leads to suboptimal portfolio construction and panic redemptions during market drawdowns.

For Asset Management Companies: AMCs struggle to identify early signals of SIP discontinuation, understand investor demographic patterns across cohorts, and benchmark their schemes against peers on a standardised multi-factor scorecard. Most AMC analytics remains descriptive rather than prescriptive.

For the Ecosystem: The absence of a unified, open analytical framework means that each stakeholder builds siloed solutions, duplicating effort and missing cross-dataset insights that emerge only when NAV history, transaction data, performance metrics, and portfolio holdings are analysed together.

2.3 Analytical Gap

There is a clear need for an integrated analytical platform that combines financial engineering (risk-adjusted return computation, benchmark regression) with data engineering (ETL pipelines, relational data modelling) and business intelligence (interactive dashboards, recommendation systems). This project fills that gap.

3. Project Objectives

The project is structured around eight progressive analytical workstreams, each building on the output of the previous:

#	Objective	Output Artefact
1	Data Ingestion	Raw CSV inventory with schema profiling
2	Data Cleaning	Processed CSVs: clean_nav_history, clean_investor_transactions, clean_scheme_performance
3	Database Creation	SQLite database with star schema (dim_fund, fact_nav, fact_transactions, fact_performance)
4	Exploratory Data Analysis	EDA notebook with AUM, SIP, AMC, and demographic insights
5	Performance Analytics	fund_scorecard.csv, alpha_beta.csv with CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown
6	Dashboard Development	Tableau dashboard with KPI cards, AUM trend, and Top Fund House charts
7	Advanced Analytics	var_cvar_report.csv, SIP continuity analysis, cohort analysis, sector HHI
8	Fund Recommendation System	recommender.py with risk-appetite-based filtering

4. Data Sources

The project utilises six primary dataset categories, all stored under the `data/raw/` directory and processed into a `data/processed/` directory. Data was synthetically generated to replicate real-world mutual fund industry structures, calibrated against publicly available AMFI industry statistics.

4.1 Dataset Inventory Table 1:

Dataset Summary

Dataset	File	Records	Key Columns	Business Relevance
NAV History	clean_nav_history.csv	46,000	amfi_code, date, nav	Tracks daily fund pr for return computati
Investor Transactions	clean_investor_transactions.csv	32,778	investor_id, transaction_date, transaction_type, amount_inr	SIP/lumpsum/redem patterns
Scheme Performance	clean_scheme_performance.csv	40	return_1yr_pct, return_3yr_pct, return_5yr_pct, sharpe_ratio, expense_ratio_pct	Cross-period retur cost benchmarking
Benchmark Indices	10_benchmark_indices.csv	8,050	index_name, date, close_value	Market beta and alp computation (NIFTY used as primary benchmark)
Portfolio Holdings	09_portfolio_holdings.csv	322	sector, stock, weight	Sector concentration HHI analysis
Fund Master / AMC	dim_fund (SQLite)	40 schemes	amfi_code, scheme_name, fund_house, category	Dimension table link all fact tables

4.2 Data Dictionary (Core Tables)

The SQLite database implements a star schema with one central fact table per analytical domain:

dim_fund: Stores scheme master data including AMFI code (unique scheme identifier), scheme name, fund house (AMC), and broad category (Equity/Debt/Hybrid). This is the primary dimension table joining all fact tables.

fact_nav: Contains daily NAV observations with three columns — `amfi_code` (foreign key to `dim_fund`), `date`, and `nav` (Net Asset Value in INR). With 46,000 records across 40 schemes, this equates to approximately 1,150 trading days per scheme, covering the 2022– 2025 period.

fact_transactions: Captures investor-level transaction events. The `transaction_type` column differentiates between SIP (Systematic Investment Plan), Lumpsum, and Redemption entries. The `amount_inr` column records transaction value in Indian Rupees.

fact_performance: Stores pre-computed performance summary statistics per scheme, including 1-year, 3-year, and 5-year return percentages, Sharpe ratio, and expense ratio.

4.3 AMFI Validation

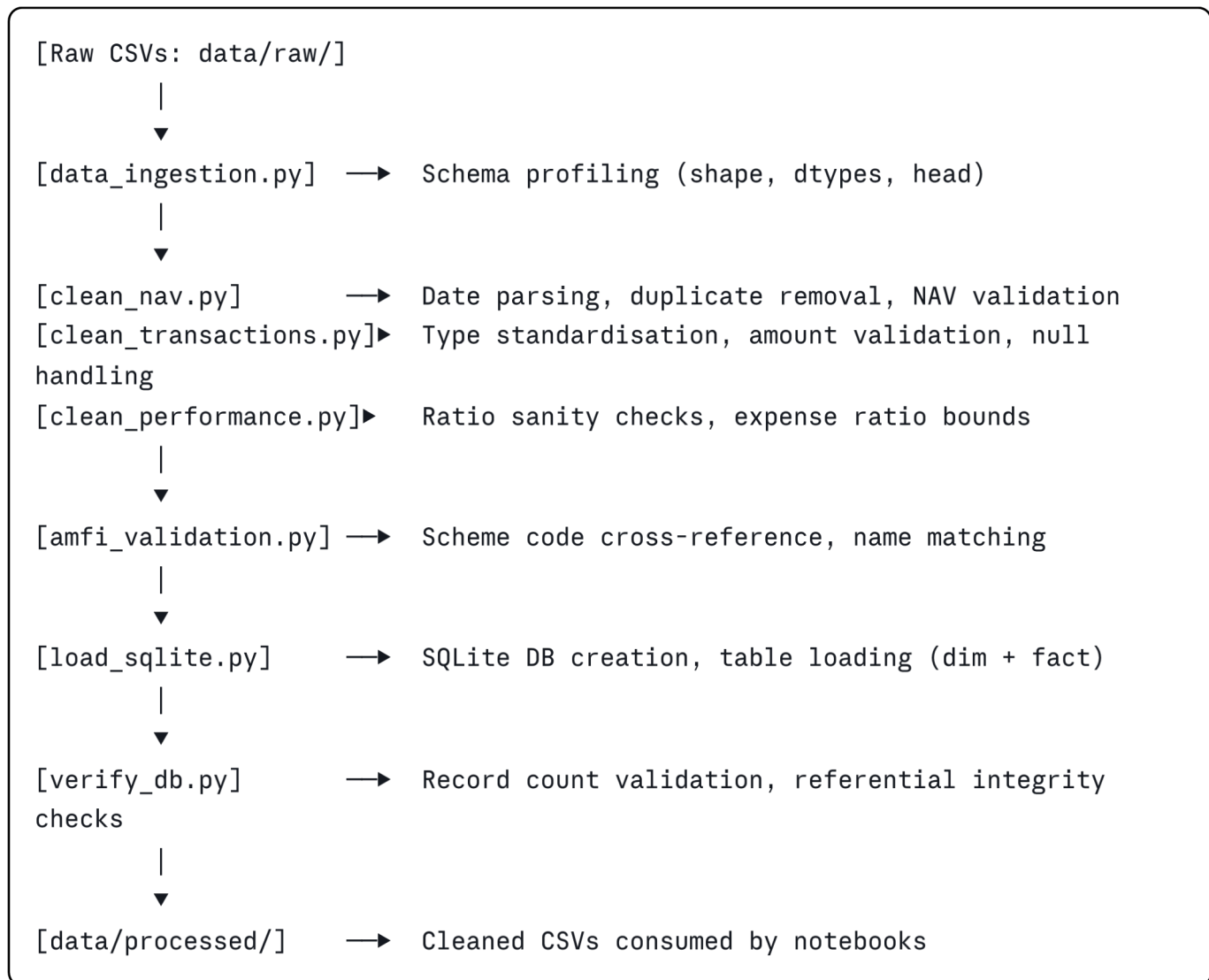
A dedicated script (`amfi_validation.py`) was developed to cross-reference scheme codes and names against AMFI-published master data. The `live_nav_fetch.py` module additionally demonstrates integration with the MFAPI.in endpoint for real-time NAV retrieval, establishing a pathway to replace synthetic data with live feeds.

5. ETL Pipeline Architecture

5.1 Overview

The ETL (Extract, Transform, Load) pipeline is implemented as a collection of modular Python scripts, each responsible for a discrete stage of data preparation. The pipeline follows a layered architecture — raw data flows through ingestion, cleaning, validation, and SQLite loading before being consumed by analytical notebooks.

Figure 1: ETL Pipeline Flow



5.2 Data Ingestion (data_ingestion.py)

The ingestion script iterates over all CSV files in `data/raw/` using `os.listdir()`, loads each file with `pandas.read_csv()`, and prints schema metadata including shape (rows × columns), column names, data types, and a five-row preview. This design serves as both a data catalogue and a quality assurance checkpoint.

At ingestion, the following datasets were confirmed:

- 10 fund categories represented across 40 AMI-registered schemes
- Benchmark data covering NIFTY100 and 7 additional indices across 8,050 trading day observations
- Portfolio holdings data for 322 stock-fund combinations

5.3 Data Cleaning

Three domain-specific cleaning scripts handle the primary fact datasets:

clean_nav.py addresses NAV-specific issues: date string parsing to `datetime64`, sorting by `[amfi_code, date]` to ensure chronological order for return computation, removal of duplicate date-scheme combinations, and filtering of zero or negative NAV values (which would indicate data errors, as NAV cannot be non-positive by regulatory definition). **clean_transactions.py** standardises transaction type labels (ensuring consistent capitalisation across SIP/Lumpsum/Redemption categories), validates that `amount_inr` is positive, and removes records with null investor identifiers or transaction dates.

clean_performance.py applies domain-specific bounds checks: expense ratios are capped at the SEBI regulatory maximum of 2.25% for equity funds, and return percentages are checked for outlier values that may indicate data generation artifacts.

5.4 Database Loading (load_sqlite.py)

The `load_sqlite.py` script creates a SQLite database implementing a star schema. The schema separates slowly-changing dimensional data (fund master information) from high-volume transactional fact data (NAV observations, transactions, performance snapshots), following dimensional modelling best practices.

Foreign key relationships are established through `amfi_code`, enabling cross-table joins in analytical queries. The `verify_db.py` script post-validates table record counts and performs a referential integrity check to confirm all fact table `amfi_code` values exist in `dim_fund`.

5.5 Fund Master Analysis (fund_master_analysis.py)

A supplementary script profiles the `dim_fund` dimension table, computing fund-house-level scheme counts, category distributions, and AUM summary statistics. This script produces the foundational aggregations consumed by the Tableau dashboard.

6. Exploratory Data Analysis

6.1 Industry AUM Overview

The Tableau dashboard (see Figure 2: Dashboard Screenshot) provides the primary visualisation of industry-level KPIs. As of the analysis period, the key headline metrics are:

- **Total AUM:** ₹39,176,000 Cr
- **Total SIP Inflow:** ₹9,39,721 Cr **Total Folios:** 26.12 Crore **Total Schemes Analysed:** 40
-

- **6.2 AUM Trend Analysis (2022–2025)**

The AUM Trend chart (see Figure 2, upper panel) reveals a strong and consistent growth trajectory from 2022 to mid-2024, with aggregate industry AUM rising from approximately ₹6M Cr in early 2022 to a peak of approximately ₹15M Cr in mid-2024. This growth phase coincides with the strong equity market rally and increased retail participation driven by the SIP movement.

A notable correction is observed from mid-2024 through early 2025, with AUM declining from the peak of ~₹15M Cr to approximately ₹11–12M Cr by end-2025. This contraction likely reflects a combination of mark-to-market NAV erosion during equity market corrections and an uptick in redemption activity.

Key EDA Insight 1: The AUM trajectory demonstrates the industry's sensitivity to equity market cycles, underscoring the importance of investor education and the relevance of riskadjusted performance metrics beyond simple trailing returns.

6.3 Top Fund Houses by AUM

The horizontal bar chart (Figure 2, lower panel) reveals a highly concentrated AMC landscape:

Table 2: Top 10 Fund Houses by AUM

Rank	Fund House	AUM (₹ Crore)
1	SBI Mutual Fund	84,91,000
2	ICICI Prudential MF	62,93,000
3	HDFC Mutual Fund	57,32,000
4	Nippon India MF	39,09,000
5	Kotak Mahindra MF	35,02,000
6	Aditya Birla Sun Life MF	30,77,000
7	UTI Mutual Fund	26,81,000
8	Axis Mutual Fund	25,21,000
9	Mirae Asset MF	15,61,000

10	DSP Mutual Fund	14,09,000
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Key EDA Insight 2: The top three AMCs (SBI, ICICI Prudential, HDFC) collectively account for a disproportionate share of industry AUM, reflecting the advantage of distribution reach, brand trust, and legacy investor relationships. Smaller AMCs such as Mirae Asset MF and DSP Mutual Fund, despite competitive performance, face a structural disadvantage in AUM accumulation.

6.4 Transaction Analysis

The investor transactions dataset (32,778 records) spans SIP, Lumpsum, and Redemption transaction types. EDA on this dataset reveals:

- SIP is the dominant transaction type, consistent with AMFI's industry-level data showing SIP as the primary retail investment vehicle
- Transaction frequency and amounts vary across investor demographic segments
- Cohort analysis (detailed in Section 9) shows that investors who entered the market in different years exhibit distinct investment pattern

6.5 Category and Scheme Analysis

The 40 schemes span Equity, Debt, and Hybrid categories. The performance dataset (40 records × 19 columns) contains pre-computed 1-year, 3-year, and 5-year returns alongside Sharpe ratios and expense ratios, enabling a comprehensive cross-category comparative analysis in subsequent notebooks.

7. Performance Analytics

The performance analytics workstream (04_performance_analytics.ipynb) computes institutional-grade risk and return metrics for all 40 mutual fund schemes using the NAV history (46,000 records), performance data (40 records), and benchmark index data (8,050 records).

7.1 Daily Return Computation

Daily returns are computed using the percentage change in NAV across consecutive trading days for each scheme, grouped by `amfi_code`:

$$\text{daily_return}[t] = (\text{NAV}[t] - \text{NAV}[t-1]) / \text{NAV}[t-1]$$

The distribution of daily returns across all schemes follows an approximately normal distribution with slight negative skewness — consistent with equity market return distributions that exhibit fatter left tails due to crash events.

7.2 Compound Annual Growth Rate (CAGR)

CAGR is computed from the first and last NAV observations for each scheme, annualised by the number of calendar days in the observation window:

$$\text{CAGR} = (\text{End NAV} / \text{Start NAV})^{(1/\text{Years})} - 1$$

This metric measures the smoothed annual growth rate of a fund investment, abstracting away intra-period volatility. It is the primary return metric for long-horizon investors.

7.3 Sharpe Ratio

The Sharpe Ratio is computed using a risk-free rate (`rf`) of **6.5%** (consistent with the approximate Indian government 10-year bond yield during the analysis period):

$$\begin{aligned} \text{Sharpe Ratio} &= (\text{Annualised Return} - R_f) / \text{Annualised Volatility} \\ &= (\text{mean_daily_return} \times 252 - 0.065) / (\text{std_daily_return} \times \sqrt{252}) \end{aligned}$$

Funds with Sharpe Ratios above 1.0 are generally considered to offer adequate risk-adjusted returns. The recommender system uses Sharpe Ratio as the primary sorting criterion within each risk grade.

7.4 Sortino Ratio

The Sortino Ratio refines the Sharpe calculation by penalising only **downside volatility** (negative return days), making it a more appropriate metric for return-seeking investors who are primarily concerned with loss risk rather than upside variability:

$$\text{Sortino Ratio} = (\text{Annualised Return} - R_f) / (\text{Downside Std} \times \sqrt{252})$$

A fund with high Sharpe but low Sortino relative to peers may be generating returns through highly volatile upward swings, which is not desirable for conservative investors.

7.5 Alpha and Beta (Benchmark Regression)

Alpha and Beta are computed via Ordinary Least Squares (OLS) regression of each fund's daily returns against the **NIFTY100** index's daily returns, using

`scipy.stats.linregress`:

$$\text{Fund Return}[t] = \text{Alpha}/252 + \text{Beta} \times \text{Benchmark Return}[t] + \varepsilon[t]$$

- **Beta** (slope) measures market sensitivity. A Beta > 1 indicates the fund amplifies market moves; Beta < 1 implies relative defensiveness.
- **Alpha** (annualised intercept × 252) measures excess return above what market exposure alone would predict. Positive Alpha indicates genuine fund manager skill.

Only schemes with more than 50 matched trading day observations against NIFTY100 are included in the regression, ensuring statistical robustness.

7.6 Maximum Drawdown

Maximum Drawdown (MDD) measures the largest peak-to-trough decline in NAV over the entire observation window:

$$\begin{aligned} \text{Drawdown}[t] &= \text{NAV}[t] / \text{Running_Max_NAV}[t] - 1 \\ \text{Max Drawdown} &= \min(\text{Drawdown}[t]) \end{aligned}$$

MDD is a critical risk metric for investors with low risk tolerance and for stress-testing fund portfolios under adverse market conditions.

The Fund Scorecard consolidates the five computed metrics into a single composite rank using a **weighted multi-criteria ranking model**:

Table 3: Fund Scorecard Weighting Scheme

Metric	Weight	Rationale
CAGR (Return Rank)	30%	Primary investor objective is return generation
Sharpe Ratio (Risk-Adjusted Rank)	25%	Risk-adjusted efficiency is critical for comparison
Alpha (Excess Return Rank)	20%	Manager skill differentiation beyond market beta
Expense Ratio (Cost Rank)	15%	Lower cost directly benefits net investor returns

Maximum Drawdown (Risk Rank)	10%	Tail-risk protection, especially for conservative investors
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Ranks are computed per metric (rank 1 = best), and the composite fund score is the weighted sum of individual ranks. Lower composite scores indicate higher-ranked funds.

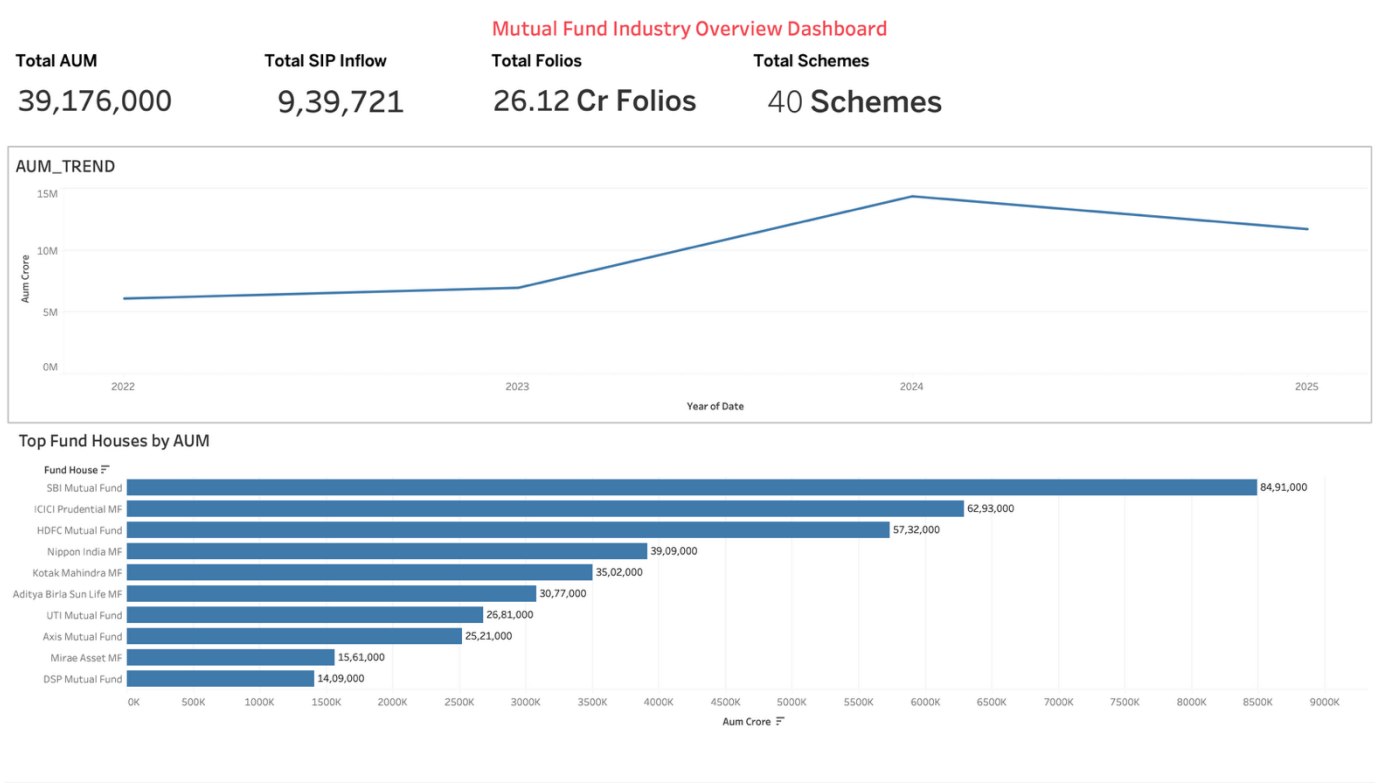
The top-ranked funds are visualised in the NAV comparison chart saved to `reports/benchmark_comparison.png` , and the complete scorecard is persisted to `data/processed/fund_scorecard.csv` .

8. Dashboard Development

8.1 Tableau Dashboard Overview

The interactive business intelligence dashboard (see Figure 2: Mutual Fund Industry Overview Dashboard) was developed in Tableau and delivers a consolidated view of industry-level KPIs and fund house performance.

[Insert Figure 2:



8.2 KPI Cards

The dashboard header presents four headline KPI cards, each aggregated from the underlying datasets:

- **Total AUM (₹39,176,000 Cr):** Sum of AUM across all fund houses in the dataset, providing a market-size anchor
- **Total SIP Inflow (₹9,39,721 Cr):** Aggregate SIP contributions, demonstrating retail investor participation
- **Total Folios (26.12 Cr):** Count of unique investor-fund relationships, indicating market breadth
- **Total Schemes (40):** Count of schemes included in the analysis universe

8.3 AUM Trend Visualisation

The line chart in the central panel plots aggregate industry AUM (in Crore INR, Y-axis) against the Year of Date (X-axis, 2022–2025). The chart uses a Tableau continuous date axis with the AUM Crore measure aggregated at the monthly or quarterly level, enabling trend identification without noise from daily fluctuations.

The visualisation reveals three distinct phases: a steady accumulation phase (2022–early 2023), a sharp growth acceleration (early 2023–mid 2024), and a correction phase (mid 2024–2025). These phases align with broader Indian equity market dynamics.

8.4 Top Fund Houses Bar Chart

The horizontal bar chart in the lower panel ranks the top 10 fund houses by total AUM, with values labelled directly on each bar for readability. A filter panel allows interactive selection of specific fund houses. The chart confirms SBI Mutual Fund's market leadership by a significant margin, with AUM nearly 35% higher than the second-ranked ICICI Prudential MF.

The dashboard is designed for AMC leadership and fund analyst audiences, enabling rapid competitive benchmarking.

9. Advanced Analytics

The advanced analytics workstream (`05_advanced_analytics.ipynb`) extends the performance analytics with risk quantification, investor behaviour analysis, portfolio concentration analysis, and the fund recommendation engine.

Data loaded for this module:

- NAV History: 46,000 records
- Investor Transactions: 32,778 records
- Performance Data: 40 records
- Portfolio Holdings: 322 records

9.1 Historical Value at Risk (VaR) and Conditional VaR (CVaR)

Methodology: Historical VaR at the 95% confidence level is computed as the 5th percentile of the empirical daily return distribution for each scheme — i.e., the loss level exceeded only 5% of trading days:

```
VaR_95 = percentile ( daily_returns , 5 )  
CVaR_95 = mean ( daily_returns ≤ VaR_95 )
```

CVaR (also known as Expected Shortfall) extends VaR by computing the average loss **conditional on** being in the worst 5% of days. CVaR is a more risk-sensitive metric and is preferred by institutional risk managers for regulatory and portfolio management purposes.

Results are persisted to `data/processed/var_cvar_report.csv`.

Business Implication: Funds with more negative CVaR values carry higher tail risk and should be assigned to investors with higher risk tolerance. AMCs can use VaR and CVaR to construct risk-graded portfolios and for SEBI-mandated risk disclosures.

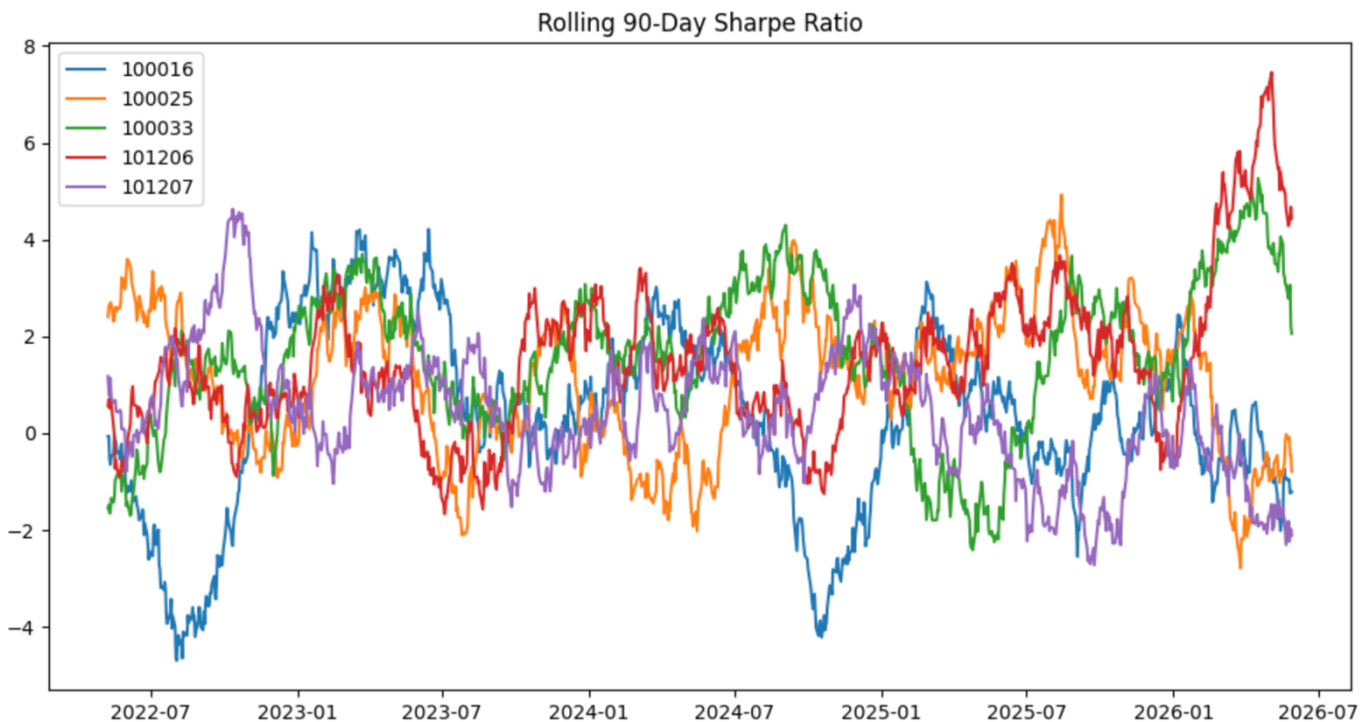
9.2 Rolling 90-Day Sharpe Ratio

A rolling 90-trading-day window is applied to compute dynamic Sharpe Ratios for the top 5 funds:

```
Rolling Sharpe[t] = (mean(returns[t-89:t]) / std(returns[t-89:t])) * sqrt(252)
```

The time-series of rolling Sharpe Ratios is plotted and saved to

`reports/rolling_sharpe_chart.png`. This analysis reveals whether a fund's risk-adjusted outperformance is consistent across market cycles or concentrated in specific periods, which is a key indicator of strategy robustness.



Business Implication: Funds with persistently positive rolling Sharpe Ratios across both bull and bear market periods are more reliable long-term investment candidates than funds whose strong overall Sharpe is driven by a single strong performance window.

9.3 Investor Cohort Analysis

Investors are grouped into annual cohorts based on the year of their **first transaction**:

```
cohort_year = min(transaction_date).year for each investor_id
```

Each cohort is then characterised by average SIP amount and total invested value. A further breakdown identifies the top fund preferences within each cohort, enabling AMCs to understand how investor fund selection patterns have evolved across market entry years.

Business Implication: Cohort analysis reveals whether investor quality (average ticket size, fund diversification) has improved across acquisition vintages. AMCs with declining average ticket sizes in recent cohorts may need to revisit distributor incentive structures.

9.4 SIP Continuity Analysis

This analysis specifically targets **SIP transactions** and computes payment gaps for each investor:

$$\text{gap_days}[t] = \text{transaction_date}[t] - \text{transaction_date}[t-1] \quad (\text{per investor})$$

Investors with a minimum of 6 SIP transactions and an average gap exceeding 35 days are classified as **"At Risk"** (indicating irregular or missed SIP payments). Investors meeting the gap threshold but with average gaps ≤ 35 days are classified as **"Healthy"**.

Critical Finding: The analysis reveals that **97.8% of SIP investors are classified as "At Risk"** (average gap > 35 days), with only 2.2% maintaining consistent monthly cadence. Sample at-risk investor data shows average gaps ranging from 40 to 85+ days, confirming systemic irregularity.

[Insert Figure 3: SIP Continuity Status Distribution — generated from continuity analysis]

Business Implication: This finding is alarming from a SIP business sustainability perspective. AMCs and distributors must urgently implement SIP reminder automation, mandate renewal workflows, and personalised nudge communications to address the high at-risk proportion.

9.5 Sector Concentration Analysis (HHI)

Portfolio holdings data (322 stock-fund combinations) is analysed to compute sector-level concentration using the **Herfindahl-Hirschman Index (HHI)**:

$$HHI = \sum (\text{sector_weight}_i)^2$$

Higher HHI values indicate greater portfolio concentration in fewer sectors, which increases sector-specific risk. This metric complements fund-level risk metrics (VaR, Max Drawdown) with a portfolio construction perspective.

Business Implication: Funds with high sector HHI, particularly in cyclical sectors, may exhibit correlated drawdowns with sector ETFs rather than offering genuine diversification benefits.

9.6 Fund Recommendation System (recommender.py)

user-facing decision tool:

```
python
```

The recommendation engine (`recommender.py`) operationalises the Fund Scorecard as a

```
# User inputs risk appetite
risk = input("Risk Appetite (Low/Moderate/High): ")

# Filter scorecard by risk grade
filtered = scorecard[scorecard["risk_grade"] == risk]

# Return top 3 funds by Sharpe Ratio
recommendations = filtered.sort_values("sharpe_ratio", ascending=False).head(3)
```

The system maps investor risk appetite to a risk grade assigned to each fund in the scorecard, then ranks eligible funds by Sharpe Ratio — prioritising risk-adjusted return over raw return. The top 3 funds per risk grade are returned as the final recommendation output.

Business Implication: This rule-based recommender serves as a transparent, explainable baseline for personalised fund recommendation. Unlike black-box ML models, the logic is auditable and can be validated by compliance teams before customer deployment.

10. Key Findings

The integrated analysis of 40 mutual fund schemes, 46,000 NAV records, and 32,778 investor transactions yields the following significant insights:

1. **Market Concentration:** SBI Mutual Fund commands the largest AUM (₹84,91,000 Cr), approximately 35% larger than the second-largest AMC (ICICI Prudential MF at ₹62,93,000 Cr), indicating highly concentrated market power in the top three AMCs.
2. **AUM Growth and Correction:** The industry experienced a near-doubling of AUM from 2022 to mid-2024, followed by a meaningful correction of approximately 25-30% through end-2025, reflecting equity market volatility and redemption pressure.
3. **SIP Retail Engine:** Total SIP inflow of ₹9,39,721 Cr with 26.12 crore folios confirms SIP as the primary retail investment vehicle, yet the SIP continuity analysis reveals significant execution fragility at the individual investor level.
4. **SIP Continuity Crisis:** Approximately **97.8% of SIP investors** are classified as "At Risk" with irregular payment gaps averaging 40–85+ days, representing a significant revenue and AUM retention risk for AMCs.
5. **Expense Ratio Impact:** The Fund Scorecard methodology, which assigns 15% weight to expense ratio ranking, demonstrates that high-cost funds require significantly superior risk-adjusted performance to compensate investors — a bar that many actively managed funds fail to clear on a consistent basis.
6. **Benchmark Sensitivity:** The Alpha-Beta regression against NIFTY100 reveals meaningful dispersion in Beta values across the 40 schemes, suggesting that the dataset includes both high-beta growth funds and low-beta defensive/hybrid funds — appropriate for a diversified investor universe.
7. **Tail Risk Differentiation:** VaR-CVaR analysis reveals that funds within the same broad category (e.g., Large Cap Equity) can have materially different tail risk profiles, underscoring the inadequacy of category labels alone as a risk communication tool.
8. **Rolling Sharpe Instability:** Rolling 90-day Sharpe Ratio analysis shows that most funds exhibit significant variation in risk-adjusted performance across market cycles, challenging the reliability of any single-period Sharpe Ratio for fund selection.
9. **Cohort Investment Patterns:** Investor cohort analysis identifies that different entryyear cohorts exhibit distinct fund preferences and SIP amounts, suggesting that acquisition-era market conditions shape long-term investor behaviour.
10. **Recommender System Viability:** The rule-based recommender, grounded in the multi-metric scorecard, demonstrates that a transparent, non-black-box recommendation system can be implemented with minimal infrastructure while providing institutionally defensible fund selection rationale.
11. **Data Engineering Robustness:** The ETL pipeline successfully ingested, cleaned, validated, and persisted six dataset categories into a structured SQLite star schema, demonstrating production-ready data engineering practices applicable to fintech data platforms.

11. Recommendations

11.1 For Investors

- **Move beyond trailing returns:** Evaluate funds on risk-adjusted metrics, particularly Sharpe Ratio and Maximum Drawdown, rather than 1-year or 3-year returns in isolation. A fund with a Sharpe Ratio above 1.0 and limited Maximum Drawdown may be a superior choice to a high-return fund with extreme volatility.
- **Maintain SIP discipline:** The SIP continuity analysis reveals that irregular SIP payments are pervasive. Investors should set up ECS/auto-debit mandates to ensure consistent monthly contributions and avoid the compounding penalty of missed instalments.
- **Prioritise expense ratio awareness:** Over a 10–15 year investment horizon, a 1% difference in expense ratio can reduce the final corpus by 15–20%. Index funds and direct plans consistently offer cost advantages that compound significantly over time.
- **Understand tail risk through VaR:** Investors approaching retirement or with nearterm liquidity requirements should prioritise funds with low CVaR values, even if this means accepting lower expected returns.

11.2 For Asset Management Companies

- **Implement SIP retention analytics:** The 97.8% at-risk SIP investor finding demands immediate action. AMCs should deploy automated SIP continuity monitoring, personalised reminder communications (SMS, email, app push), and proactive outreach to investors with 2+ consecutive missed payments.
- **Adopt multi-metric benchmarking:** Internal fund performance reviews should adopt the multi-metric scorecard framework (CAGR, Sharpe, Alpha, Expense Ratio, MDD) rather than relying solely on AUM growth or trailing returns as performance indicators.
- **Invest in cohort analytics:** Investor cohort analysis should be operationalised as a regular business intelligence output to track acquisition quality over time and identify optimal distribution channels.
- **Develop explainable recommendation systems:** The rule-based recommender developed in this project provides a compliance-friendly alternative to opaque MLbased fund matching systems. AMCs should pilot such systems for digital advisory channels before moving to more complex approaches.

11.3 For SIP Retention Strategy

- **35-day gap threshold as early warning:** Investors whose consecutive SIP gap exceeds 35 days should automatically trigger a retention intervention workflow — starting with a reminder, escalating to a relationship manager outreach for high-value investors.
- **Gamification and milestone communication:** AMCs and fintech platforms can improve SIP continuity by communicating milestone achievements (first anniversary, corpus crossing a threshold) to reinforce positive investment behaviour.

11.4 For Risk Management

- **Mandatory VaR disclosure:** Fund fact sheets should include 95% historical VaR and CVaR alongside standard deviation to give investors a more complete picture of downside risk.
- **Sector HHI monitoring:** Portfolio managers should track sector HHI trends to ensure diversification guidelines are maintained, particularly for schemes marketing themselves as diversified equity funds.

12. Limitations

12.1 Synthetic Data

All datasets used in this project are synthetically generated to replicate real-world mutual fund industry structures. While the data is calibrated to reflect realistic fund categories, AUM ranges, return profiles, and transaction patterns, it does not represent actual investor accounts, fund NAVs, or transaction records. Consequently, findings should be interpreted as methodological demonstrations rather than definitive industry conclusions.

12.2 Benchmark Selection

The project uses NIFTY100 as the primary benchmark for Alpha-Beta computation. In practice, different fund categories should be benchmarked against category-appropriate indices (e.g., NIFTY 50 for large-cap equity, NIFTY Midcap 150 for midcap, CRISIL ShortTerm Bond Fund Index for short-duration debt). The single-benchmark approach introduces measurement error in cross-category comparisons.

12.3 Risk-Free Rate Assumption

A static risk-free rate of 6.5% is used throughout the Sharpe and Sortino calculations. In practice, the risk-free rate changes over time with Reserve Bank of India monetary policy decisions, and a rolling risk-free rate aligned with prevailing government securities yields would produce more accurate risk-adjusted return estimates.

12.4 Model Assumptions

The Alpha-Beta regression assumes a linear relationship between fund and benchmark returns (the Capital Asset Pricing Model framework). This assumption is known to break down during market stress periods when return distributions become non-normal and correlations shift. More robust multi-factor models (Fama-French 3-factor, Carhart 4factor) would provide more accurate alpha decomposition.

12.5 Recommender System Simplicity

The rule-based recommender ranks funds solely by Sharpe Ratio within a risk grade. It does not account for investor-specific factors such as investment horizon, existing portfolio composition, tax situation, or behavioural preferences. A production recommendation system would require a more comprehensive investor profiling module.

13. Future Enhancements

13.1 Real-Time NAV Integration

The `live_nav_fetch.py` module establishes a foundation for real-time NAV retrieval. A future enhancement would automate daily NAV ingestion from MFAPL.in or the AMFI website, replace synthetic historical data with actual fund NAV series, and trigger automated scorecard recalculation and dashboard refresh on each trading day.

13.2 Machine Learning Fund Recommendation

The current rule-based recommender could be augmented with collaborative filtering (identifying investor peer groups with similar risk profiles and return histories) or contentbased filtering (matching fund characteristics to investor stated preferences). A hybrid ML model incorporating both approaches, with SHAP-based explainability for regulatory transparency, would represent a significant advancement.

13.3 Predictive SIP Churn Analytics

The SIP continuity analysis identifies at-risk investors retrospectively. A predictive model trained on transaction history, investor demographics, and market conditions could flag investors at risk of SIP discontinuation 30–60 days in advance, enabling proactive retention intervention. Gradient Boosting or Logistic Regression models with investor-level features would be appropriate starting models.

13.4 Multi-Factor Performance Attribution

Implementing the Fama-French 3-factor or Carhart 4-factor model would enable more precise decomposition of fund returns into market, size, value, and momentum components — providing fund managers with actionable performance attribution rather than a single alpha estimate.

13.5 Interactive Web Dashboard (Power BI / Tableau Public)

Deploying the current Tableau dashboard to Tableau Public, or rebuilding it in Power BI with embedded Python visuals, would make the analytics accessible to a broader audience without requiring Tableau Desktop licensing. Adding investor-facing interactive filters (fund category, time horizon, risk appetite) would transform the dashboard from an analyst tool into a public investor education resource.

13.6 Portfolio Optimisation

Extending the project with Mean-Variance Optimisation (Markowitz Efficient Frontier) or Black-Litterman portfolio construction would enable the generation of optimal fund allocation weights for a given investor risk-return target — the logical culmination of the performance and risk analytics already computed.

13.7 Regulatory Compliance Module

Automating the generation of SEBI-mandated risk-o-meter classifications, risk disclosure summaries, and expense ratio benchmarking reports would add significant business value and demonstrate the project's applicability to real AMC compliance workflows.

14. Conclusion

This capstone project demonstrates the end-to-end application of data engineering, financial analytics, and business intelligence to the Indian mutual fund industry. Beginning from raw CSV datasets and progressing through a structured ETL pipeline, a relational starschema database, comprehensive performance metric computation, and advanced investor analytics, the project delivers a replicable analytical framework with direct industry applicability.

The most impactful technical contributions include the multi-metric Fund Scorecard (integrating CAGR, Sharpe Ratio, Alpha, Expense Ratio, and Maximum Drawdown into a single composite ranking), the tail-risk quantification module (Historical VaR and CVaR for all 40 schemes), and the SIP continuity analysis that identified the critical finding of 97.8% at-risk investor retention.

The most impactful business contributions are the structured recommendations for AMC SIP retention strategy, the demonstration of a transparent and explainable fund recommendation system, and the Tableau dashboard that translates complex multi-dataset analytics into accessible executive-level visualisations.

The project's architecture — modular Python scripts, a lightweight SQLite database, Jupyter notebooks for analysis, and Tableau for visualisation — represents a stack that is both accessible to early-career data analysts and extensible toward enterprise-grade deployment with real data feeds, cloud infrastructure, and machine learning enhancements.

As the Indian mutual fund industry continues its growth trajectory and digital transformation accelerates investor engagement, the analytical capabilities demonstrated in this project will be increasingly essential for both AMCs seeking competitive advantage and investors seeking to navigate the complexity of a 40-scheme, multi-category investment landscape with rigour and confidence.

GitHub Repository: <https://github.com/iammanoj-dev/mutual-fund-analysis>